(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

VI Semester, March 2025

Paper Code: DA-304T **Sub.: Statistics, Statistical Modelling & Data Analytics**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No. 1

i. What are discrete and continuous variables? 2 (CO1)

ii. Give the relationship between mean, median and mode for positively skewed, negatively skewed and symmetric distributions. 2 (CO1)

iii. Give mathematical representations for simple and multi-linear regression models. 2 (CO2)

iv. Write properties of linear estimators. 2 (CO2)

v. State the assumptions made for unobservable errors in the regression models of Q. 1 (iii). 2 (CO2)

Q. No. 2

Explain in detail sampling distribution of the means. What is the significance of Central limit theorem in ascertaining the nature of such distributions? Hence, discuss sampling statistics for (a) sampling with and without replacement, and (b) large and small sampling cases. 10 (CO1)

Q. No. 3

i. (a) Find the pdf for the standard normal distribution. 2 (CO1)
(b) Explain the significance of chi-square distribution in statistics, and F-distribution in model (variable) selection. 3 (CO1, CO2)

ii. Two distances are measured as 27.3 centimeters (cm) and 15.6 cm with standard deviations (standard errors) of 0.16 cm and 0.08 cm, respectively. Determine the mean and standard deviation of (a) the sum and (b) the difference of the distances. 5 (CO1)

Q. No. 4

i. Explain least-squares method for estimation of regression parameters, and derive expressions for linear estimators. 5 (CO2)

ii. Write short notes on *any two* - 2x2.5 =5 (CO2)
(a) Gauss-Markov Theorem (b) Factorial Experiments
(c) Logistic regression model (d) Transformations & Box-Cox model